

Research Proposal

Agentic Retrieval-Augmented Generation (RAG) for Intelligent Investment Analysis and Financial Decision Support

Title

Agentic Retrieval-Augmented Generation (RAG) Framework for Intelligent Investment Analysis, Financial Forecasting, and Autonomous Decision Support

Abstract

The rapid growth of Large Language Models (LLMs) has transformed financial analytics, enabling advanced natural language understanding and automated reasoning. However, traditional LLMs suffer from limitations such as hallucinations, outdated knowledge, and insufficient domain-specific reasoning capabilities. Retrieval-Augmented Generation (RAG) addresses these limitations by integrating external knowledge retrieval with generative AI systems. More recently, Agentic RAG architectures have emerged, combining autonomous agents, iterative reasoning, dynamic retrieval, and tool execution to create intelligent financial decision-support systems.

This research proposes the design and implementation of an Agentic RAG framework for investment analysis capable of autonomously retrieving financial documents, interpreting market conditions, analyzing company fundamentals, forecasting financial performance, and generating explainable investment recommendations. The proposed framework integrates vector databases, multi-agent systems, financial knowledge graphs, and reinforcement learning-based reasoning strategies.

The study aims to evaluate whether Agentic RAG systems outperform conventional financial analytics methods and standard RAG pipelines in terms of accuracy, explainability, adaptability, and real-time decision-making. Experimental validation will be conducted using financial datasets including SEC filings, earnings reports, macroeconomic indicators, and stock market data from the U.S. equity market.

1. Introduction

Artificial Intelligence has become increasingly influential in financial markets, investment banking, portfolio optimization, and risk analysis. Modern financial institutions process enormous volumes of structured and unstructured information including:

- SEC filings
- Earnings call transcripts
- News articles
- Macroeconomic reports
- Market sentiment
- Financial statements

Traditional machine learning models often struggle with the contextual and dynamic nature of financial information. Large Language Models (LLMs) provide substantial improvements in textual understanding; however, they remain constrained by static training knowledge and hallucination risks.

Retrieval-Augmented Generation (RAG) introduces a hybrid paradigm where external financial documents are dynamically retrieved before answer generation. Agentic RAG extends this architecture further by incorporating autonomous agents capable of:

- multi-step reasoning,
- iterative retrieval,
- tool usage,
- planning,
- memory management,
- self-evaluation,
- and adaptive investment analysis.

This proposal investigates how Agentic RAG systems can be used to develop intelligent investment analysis platforms capable of supporting financial analysts, portfolio managers, and institutional investors.

2. Research Problem

Current AI-based investment analysis systems face several limitations:

1. Hallucinated financial interpretations
2. Lack of real-time market awareness
3. Inability to perform autonomous reasoning
4. Poor explainability in financial recommendations
5. Difficulty integrating heterogeneous financial data
6. Limited capability for dynamic market adaptation

Traditional RAG systems retrieve information but generally lack strategic reasoning and autonomous planning capabilities.

The research problem is therefore:

How can Agentic RAG architectures improve financial investment analysis through autonomous retrieval, reasoning, and explainable decision-making?

3. Research Objectives

General Objective

To develop and evaluate an Agentic Retrieval-Augmented Generation framework for autonomous investment analysis and financial decision support.

Specific Objectives

1. Design an Agentic RAG architecture specialized for financial analysis.
2. Integrate financial knowledge retrieval from:
 - SEC filings,
 - market data,
 - macroeconomic reports,
 - and financial news.
3. Develop autonomous financial agents capable of:
 - reasoning,

- planning,
 - iterative retrieval,
 - and investment recommendation generation.
4. Implement vector-based semantic retrieval using financial embeddings.
 5. Compare Agentic RAG against:
 - standard RAG,
 - transformer-only models,
 - and traditional ML forecasting methods.
 6. Evaluate explainability, accuracy, and forecasting performance.
-

4. Research Questions

1. Can Agentic RAG systems improve investment decision quality compared to traditional financial AI systems?
 2. How effectively can autonomous agents perform iterative financial reasoning?
 3. Does integrating real-time retrieval reduce hallucinations in financial analysis?
 4. Can Agentic RAG improve explainability in investment recommendations?
 5. What financial indicators contribute most to autonomous investment forecasting?
-

5. Literature Review

5.1 Retrieval-Augmented Generation (RAG)

RAG combines retrieval systems with generative AI models.

Basic workflow:

Query → Retriever → Relevant Documents → LLM → Response

Lewis et al. (2020) introduced RAG as a method to improve factual consistency in language generation.

Applications in finance include:

- financial report analysis,

- risk management,
 - and automated compliance systems.
-

5.2 Agentic AI Systems

Agentic AI systems perform:

- autonomous planning,
- reasoning,
- self-reflection,
- and tool execution.

Recent frameworks include:

- AutoGPT,
- LangGraph,
- CrewAI,
- and ReAct architectures.

Agentic systems enable iterative decision-making rather than single-pass generation.

5.3 AI in Investment Analysis

AI has been used for:

- algorithmic trading,
- portfolio optimization,
- fraud detection,
- and sentiment analysis.

Machine learning methods include:

- Random Forest,
- XGBoost,
- LSTM,

- and Reinforcement Learning.

However, many systems lack explainability and contextual reasoning.

5.4 Financial Knowledge Retrieval

Financial retrieval systems commonly use:

- embeddings,
- vector databases,
- semantic search,
- and hybrid retrieval.

Popular vector databases:

- Pinecone,
- FAISS,
- Weaviate,
- Milvus.

6. Proposed Methodology

6.1 Proposed Architecture

The proposed Agentic RAG framework contains:

A. Financial Data Layer

Sources include:

- SEC EDGAR
 - Yahoo Finance
 - Alpha Vantage
 - Bloomberg datasets
 - Financial news APIs
-

B. Embedding Layer

Financial documents will be transformed into vector embeddings.

Document representation:

C. Vector Retrieval Layer

Similarity retrieval will use cosine similarity:

D. Agentic Reasoning Layer

The agent system performs:

- query decomposition,
 - iterative retrieval,
 - financial reasoning,
 - verification,
 - and recommendation synthesis.
-

E. Financial Forecasting Layer

Time-series prediction models:

- LSTM,
 - Transformer forecasting,
 - ARIMA,
 - Reinforcement Learning.
-

7. Proposed Multi-Agent Framework

Agent	Function
Retrieval Agent	Retrieve financial documents

Agent	Function
Market Agent	Analyze market conditions
Risk Agent	Estimate investment risk
Forecasting Agent	Predict future performance
Portfolio Agent	Recommend allocation
Verification Agent	Validate outputs

8. Experimental Design

Dataset

The study will use:

- S&P 500 companies
 - SEC 10-K filings
 - Earnings transcripts
 - Historical stock prices
 - Economic indicators
-

Evaluation Metrics

Financial Forecasting Metrics

Mean Squared Error:

Root Mean Squared Error:

NLP Evaluation

- Retrieval Precision
- Recall
- Faithfulness

- Hallucination Rate
 - Explainability Score
-

9. Expected Contributions

This research is expected to contribute:

1. A novel Agentic RAG architecture for finance
 2. Autonomous financial reasoning mechanisms
 3. Improved explainability in investment recommendations
 4. Enhanced real-time financial intelligence systems
 5. A scalable framework for institutional financial analytics
-

10. Expected Results

The proposed system is expected to:

- outperform standard RAG systems,
 - reduce hallucinations,
 - improve retrieval precision,
 - provide explainable investment insights,
 - and improve financial forecasting accuracy.
-

11. Significance of the Study

The study may significantly impact:

- fintech,
- investment banking,
- hedge funds,
- portfolio management,
- robo-advisors,

- and institutional AI analytics.

The framework could support:

- autonomous market intelligence,
 - explainable AI,
 - and next-generation financial assistants.
-

12. Future Extensions

Future work may include:

- multimodal RAG with charts and tables,
 - graph neural networks,
 - reinforcement learning trading agents,
 - decentralized financial AI systems,
 - and quantum-enhanced portfolio optimization.
-

13. Proposed Tools and Technologies

Component Technology

LLM GPT / Llama / Claude

Vector DB Pinecone / FAISS

Framework LangChain / LlamaIndex

Forecasting PyTorch / TensorFlow

Agents CrewAI / LangGraph

Data Sources SEC EDGAR / Yahoo Finance

14. Timeline

Phase	Duration
Literature Review	2 Months
Data Collection	1 Month
System Design	2 Months
Implementation	3 Months
Experimental Analysis	2 Months
Writing & Publication	2 Months

15. References

1. Lewis, P. et al. (2020). *Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks*.
2. OpenAI (2025). *Agentic AI Systems and Autonomous Reasoning*.
3. CFA Institute. *Retrieval-Augmented Generation in Finance*.
4. Vaswani, A. et al. (2017). *Attention is All You Need*.
5. Hochreiter, S. & Schmidhuber, J. (1997). *Long Short-Term Memory*.
6. ReAct Framework (2023). *Reasoning and Acting in Language Models*.
7. LangChain Documentation.
8. SEC EDGAR Database Documentation.
9. Financial NLP Research Surveys (2024–2025).
10. Recent studies on multimodal RAG and financial AI systems.